

CS3001 – Computer Networks
Quiz – 2B

Name	
Roll. No.	
Section	

Part 1 (8 Marks)

Suppose an enterprise network has the following parameters:

- **Access link rate = 3 Mbps**
- **Average object size = 300 Kbits**
- **Average request rate from all browsers = 8 requests/sec**
- **RTT to the origin server = 2.0 sec and LAN delay is negligible (0ms).**

With local web cache: Cache hit rate = **60%** (all other requests go to the origin servers). Ignore queueing when utilization ≤ 0.8 but assume **queueing delays infinite when utilization > 0.9 .**

	Without Cache	With Cache
Average access link utilization		
Average request delay		

Delay = Internet delay (RTT) + Access link transmission delay.

Part 2 (2 Marks)

You are tasked with designing an application-layer protocol for a **real-time stock trading platform** where brokers and clients interact continuously. The protocol must support:

- Millisecond-level latency for transaction updates.
 - Occasional large message bursts (market data feeds).
 - Strong encryption and authentication of all sessions.
 - Reliability is critical **only for trades and account commands**, but **market data feeds can tolerate some loss**.
1. Which transport protocol (**TCP, UDP, UDP with TLS, TCP with TLS**, or a **combination (hybrid)** of protocols) would you use and why?
 2. How should the state be maintained between client and server sessions (cookies vs server-managed state vs stateless) and why?

	Protocol(s)	Why?
1.		
2.		

Part 1 Solution:

Step 1: Calculate traffic intensity (no cache).

- Arrival rate = 8 requests/sec
- Object size = 300 Kbits
- Data rate required = $8 \times 300 = 2400$ Kbps = 2.4 Mbps

Access link capacity = **3 Mbps** → Utilization:

$$U = 2.4/3 = 0.8$$

This is exactly at the threshold → **stable (no heavy queueing yet)**.

Step 2: With cache hit rate = 0.6.

- 40% of requests go to the origin server.
- Effective arrival rate = $0.4 \times 8 = 3.2$ req/sec
- Data rate = $3.2 \times 300 = 960$ Kbps

Utilization:

$$U = 960/3000 = 0.32$$

This is well below 0.8 → **safe (no queueing buildup)**.

Step 3: Expected delay (formula applied).

- Transmission delay = $300/3000 = 0.1$ s
- **Without cache:**
 - RTT = 2.0 s
 - Transmission = 0.1 s
 - Total delay per request = **2.1 s**
 - Utilization = 0.8 → acceptable, but close to the limit.
- **With cache:**
 - Cache hit delay = 0 (LAN negligible)
 - Cache miss delay = 2.1 s
 - Average delay = $(0.6 \times 0) + (0.4 \times 2.1)$

$$E[\text{Delay}] = 0.84 \text{ s}$$

Final Answer:

- **Without cache:**
 - Utilization = 0.8 (stable, just at limit)
 - Average delay = **2.1 s**
- **With cache:**
 - Utilization = 0.32 (very safe)
 - Average delay = $\approx$ **0.84 s**

→ **Both scenarios are technically acceptable, but caching reduces delay significantly and ensures stability with much lower utilization.**

Part 2 Solution:

1. Transport protocol:

- **TCP:** Provides reliability, but retransmission delays make it unsuitable for real-time market data feeds.
- **UDP:** Low-latency, suitable for real-time feeds, but unreliable.
- **Hybrid design:**
 - **TCP (with TLS)** for critical trades and command messages requiring guaranteed delivery.
 - **UDP with TLS** for streaming market data feeds, where some packet loss is acceptable but encryption/authentication is required.

Answer: A **hybrid approach**: TCP+TLS for control/trade-critical data, **UDP with TLS** for live market feeds.

2. State maintenance:

- Stateless operation is not possible since client accounts and positions must persist.
- Cookies are insecure and insufficient for financial transactions.
- Best choice → **server-managed state**, where each client session ties to an authenticated account object. This ensures persistence, even if the client temporarily disconnects.